METADATA

```
┌──(hafizi㉿kali)-[~]
└─$ cd ~/Desktop
mkdir metadata_lab
cd metadata_lab

┌──(hafizi㉿kali)-[~/Desktop/metadata_lab]
└─$ 
```

1. Ocean.jpg

```
┌──(hafizi㉿kali)-[~/Desktop/metadata_lab]
└─$ exiftool ocean.jpg
ExifTool Version Number         : 13.50
File Name                       : ocean.jpg
Directory                       : .
File Size                       : 43 kB
File Modification Date/Time     : 2026:04:09 05:27:39-04:00
File Access Date/Time           : 2026:04:09 05:27:59-04:00
File Inode Change Date/Time     : 2026:04:09 05:27:58-04:00
File Permissions                : -rw-rw-rw-
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : inches
X Resolution                    : 72
Y Resolution                    : 72
Profile CMM Type                : Little CMS
Profile Version                 : 2.1.0
Profile Class                   : Display Device Profile
Color Space Data                : RGB
Profile Connection Space        : XYZ
Profile Date Time               : 2012:01:25 03:41:57
Profile File Signature          : acsp
Primary Platform                : Apple Computer Inc.
CMM Flags                       : Not Embedded, Independent
Device Manufacturer             :
Device Model                    :
Device Attributes               : Reflective, Glossy, Positive, Color
Rendering Intent                : Perceptual
Connection Space Illuminant     : 0.9642 1 0.82491
Profile Creator                 : Little CMS
Profile ID                      : 0
Profile Description             : c2
Profile Copyright               : IX
Media White Point               : 0.9642 1 0.82491
Media Black Point               : 0.01205 0.0125 0.01031
Red Matrix Column               : 0.43607 0.22249 0.01392
Green Matrix Column             : 0.38515 0.71687 0.09708
Blue Matrix Column              : 0.14307 0.06061 0.7141
Red Tone Reproduction Curve     : (Binary data 64 bytes, use -b option to extract)
Green Tone Reproduction Curve   : (Binary data 64 bytes, use -b option to extract)
Blue Tone Reproduction Curve    : (Binary data 64 bytes, use -b option to extract)
Comment                         : THIS IS THE HIDDEN FLAG
Image Width                     : 640
Image Height                    : 425
Encoding Process                : Progressive DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 640x425
Megapixels                      : 0.272
```

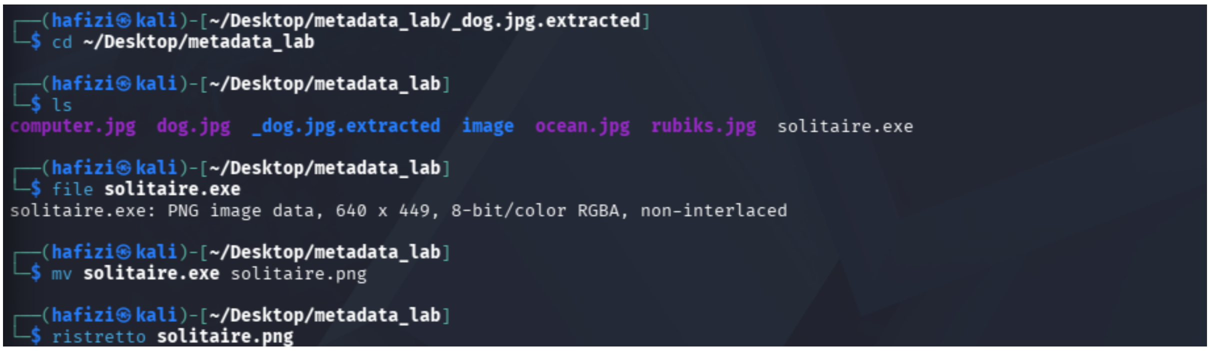
The file **ocean.jpg** was examined using the exiftool command to extract its metadata. The analysis revealed a **Comment field** containing the hidden message "THIS IS THE HIDDEN FLAG". This shows that sensitive information can be stored within image metadata, a form of **metadata leakage**, without affecting the visible content of the file.

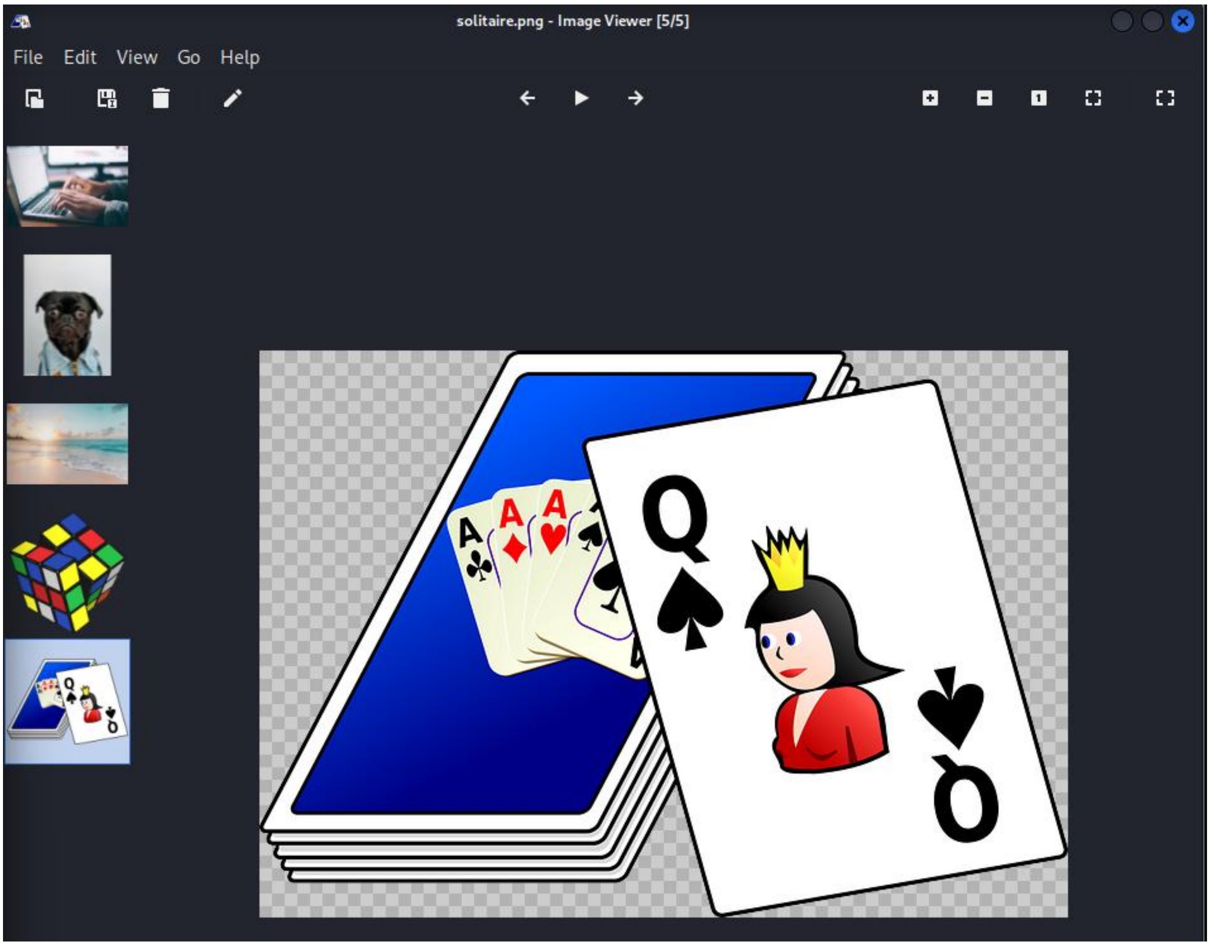
2. dog.jpg

```
┌──(hafizi㉿kali)-[~/Desktop/metadata_lab]
└─$ binwalk dog.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             JPEG image data, JFIF standard 1.01
88221         0x1589D         Zip archive data, at least v1.0 to extract, compressed size: 21, uncompressed size: 2
1, name: hidden_text.txt
88384         0x15940         End of Zip archive, footer length: 22

┌──(hafizi㉿kali)-[~/Desktop/metadata_lab]
└─$ binwalk -e dog.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
88221         0x1589D         Zip archive data, at least v1.0 to extract, compressed size: 21, uncompressed size: 2
1, name: hidden_text.txt

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

┌──(hafizi㉿kali)-[~/Desktop/metadata_lab]
└─$ cd _dog.jpg.extracted
ls
1589D.zip  hidden_text.txt

┌──(hafizi㉿kali)-[~/Desktop/metadata_lab/_dog.jpg.extracted]
└─$ cat hidden_text.txt
THIS IS A HIDDEN FLAG
```

The file **dog.jpg** was analyzed using the binwalk tool to detect embedded data. The scan identified a hidden ZIP archive containing a file named **hidden_text.txt**. After extracting the contents using binwalk -e, the hidden file was accessed and revealed the message "THIS IS A HIDDEN FLAG". This demonstrates the concept of **file embedding**, where additional data is concealed within another file.

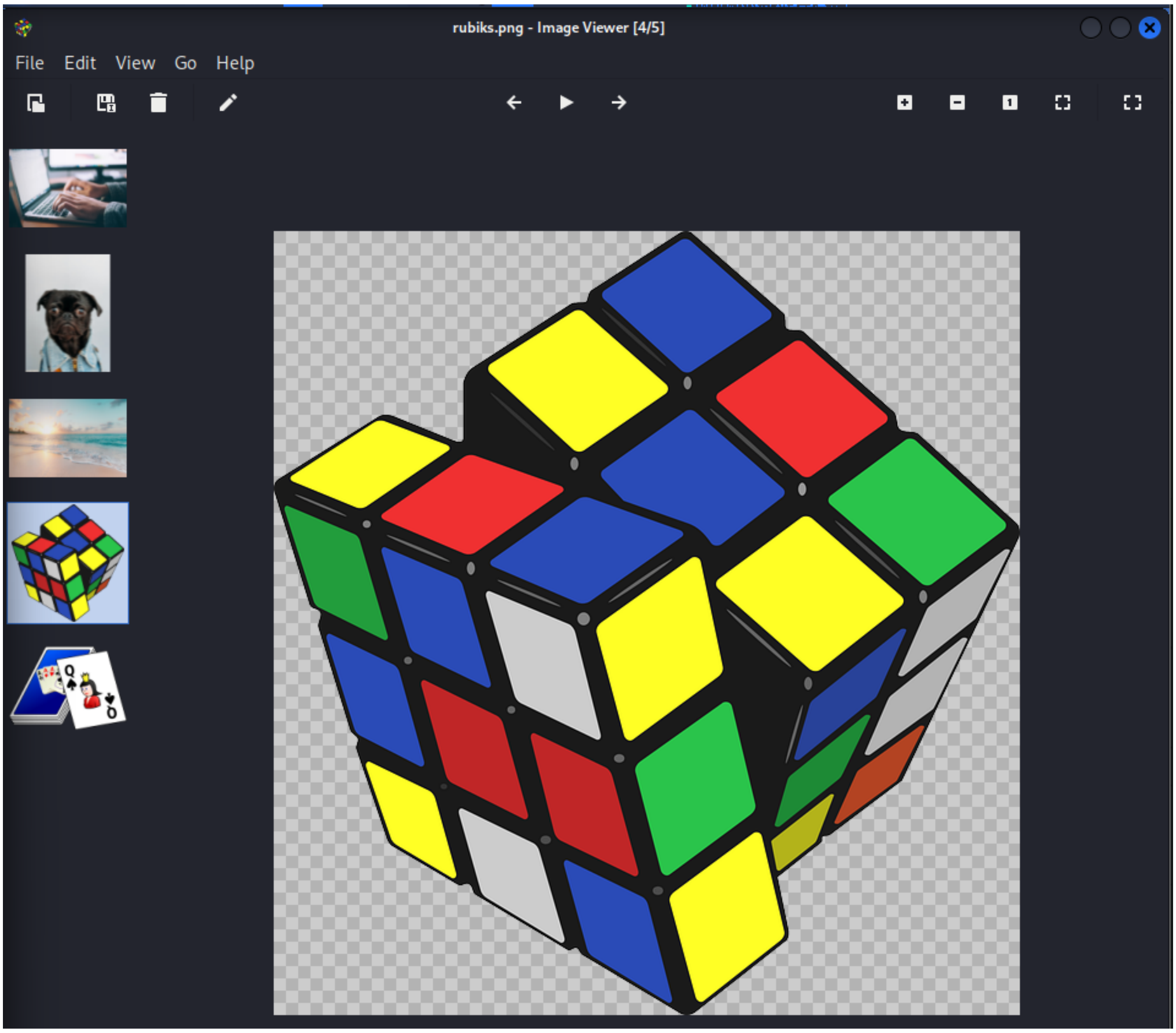
3. solitaire.exe





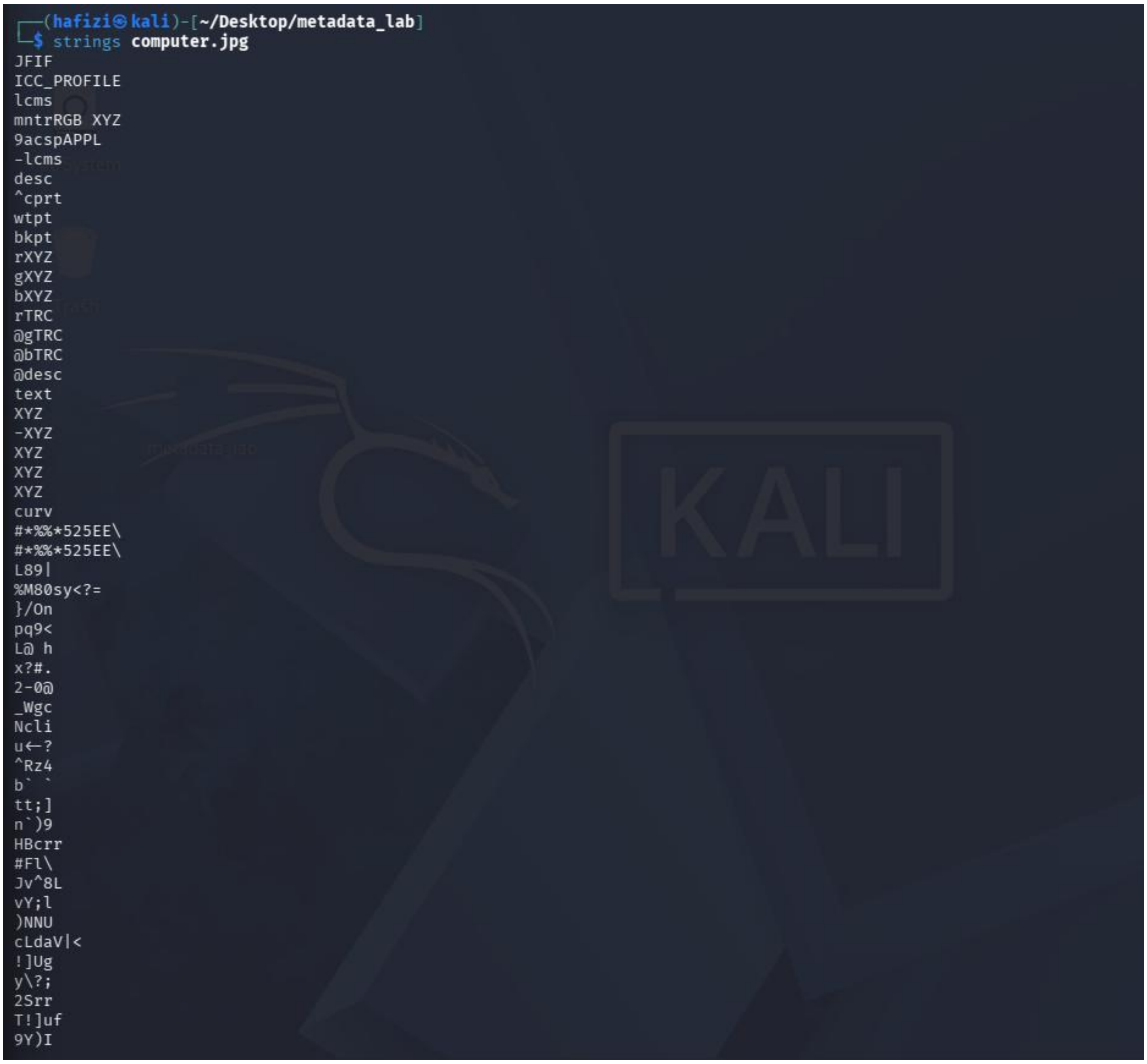
The file **solitaire.exe** was checked using the file command to determine its actual file type. The result showed that it was not an executable, but a **PNG image (640 x 449, RGBA)**. This confirms that the file extension was misleading. After renaming it to **solitaire.png**, the file opened successfully as an image. This highlights the concept of **file extension spoofing**, where file names are manipulated to disguise their true nature.

4. rubiks.jpg



The file **rubiks.jpg** was analyzed using the file command to verify its actual format. The output indicated that the file was a **PNG image (609 x 640, RGBA)** instead of a JPEG. After correcting the extension to **rubiks.png**, the image opened successfully. This is another example of **file extension spoofing**, demonstrating how attackers may disguise file types to mislead users.

5. computer.jpg



The file **computer.jpg** was analyzed using the strings command to extract readable data from its binary content. Several meaningful strings were identified, including **"ICC_PROFILE"**, **"mntrRGB XYZ"**, and **"JFIF"**, which relate to the image's internal structure and color profile. This demonstrates how **string extraction** can reveal hidden or embedded information within binary files.

IN CONCLUSION

In conclusion, this lab highlights how various file analysis techniques can uncover hidden information that is not visible through normal inspection. Methods such as metadata extraction, embedded file detection, file signature verification, and string analysis reveal potential security risks such as metadata leakage, file embedding, and file extension spoofing. These techniques are essential in cybersecurity, as they help identify concealed data and detect suspicious or manipulated files.